

Shouyu Lai

Ph.D. Student in Artificial Intelligence | Robotics, Reinforcement Learning, and Control

4573 Laboratory, Beihang University, China | laishouyu078@gmail.com | github.com/1aStlY

PROFILE

Robotics Ph.D. student working at the intersection of robot control, reinforcement learning, whole-body control, mechanical design, and real-world robotic systems. Interested in robust sim-to-real deployment and generalizable control across humanoids, quadrupeds, UAVs, and reconfigurable robots.

EDUCATION

2024-Present	Beihang University	Ph.D. Student, Artificial Intelligence (transfer from master's program)
2022-2024	Beihang University	Master's studies, Mechanical Engineering and Automation
2018-2022	Harbin Engineering University	B.Eng., Mechanical Engineering and Automation

RESEARCH INTERESTS

Robot control; reinforcement learning; whole-body control; legged robotics; sim-to-real; aerial robotics; mechanical design; serial-parallel hybrid mechanisms; reconfigurable robots; mechatronic system integration.

SELECTED PROJECTS

Quadruped Manipulator: Reinforcement Learning + Whole-Body Control

Simulation-to-real workflow across Isaac Lab, MuJoCo, and a real quadruped manipulator platform.

Serial-Parallel Hybrid Reconfigurable Robot

Mechanism and prototype design for morphology reconfiguration using a serial-parallel hybrid architecture.

Autonomous UAV and Anti-UAV Platforms

Integrated platforms including multi-camera/LiDAR UAVs, gimbal/LiDAR UAVs, net-capture systems, and signal-suppression systems.

Compact Aerial Robot System Engineering

Integration of flight control, propulsion, LiDAR, edge AI computing, power, communication, and mechanical packaging.

TECHNICAL SKILLS

Learning & Control	Reinforcement learning, robot control, whole-body control, sim-to-real, policy deployment
Simulation & Software	Python, PyTorch, Isaac Lab, MuJoCo, robot simulation and data pipelines
Mechanical Engineering	Mechanical design, CAD and prototyping, mechanisms, reconfigurable systems
Robot Systems	UAV integration, flight control, Orin NX, LiDAR, power architecture, mesh communication

PUBLICATIONS

No public papers listed yet. Current work focuses on building, validating, and documenting real robotic systems.